

WILLEM SMITH

Full-Stack Developer — Enterprise Systems & AI-Driven Engineering

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PROFILE

Full-stack software developer with a BSc in Information Technology. Currently a Full-Stack Developer at Thoughtware (Pty) Ltd, where I independently own major feature areas of a production POS platform end-to-end — React Native frontend, FastAPI backend, and MSSQL data layer — with a focus on real-time systems, API design, and reliability engineering.

KEY ACHIEVEMENTS

- Solo full-stack developer across major areas of a production POS platform, spanning React Native, FastAPI, and MSSQL.
- Designed a two-phase commit and reconciliation job that closed a data-loss window in live payment processing.
- Built a real-time WebSocket event bus broadcasting live order, table, and staff updates across every connected POS terminal.
- Designed and implemented an AI context infrastructure spanning 8 integrated applications and ~90 API routes — reducing token costs and hallucinations in AI-assisted development.

WORK EXPERIENCE

Full-Stack Developer — Thoughtware (Pty) Ltd

April 2026 – Present | Johannesburg, South Africa

React Native • React • TypeScript • Vite • Python • FastAPI • SQLAlchemy • MSSQL • WebSockets • JWT • AsyncStorage • ESC/POS

Designed and built a point-of-sale ecosystem from scratch — architected and led development of the core POS app and 4 companion apps: Customer Display System, Manager Dashboard, and Back Office built from scratch, and the Kitchen Display System extended with new capabilities, over ~4.5 months.

- Built the core POS application solo: order/payment flow, floor plan and table management, staff/PIN permissions, offline-resilient sync, and a full first-run setup wizard — the primary app the rest of the ecosystem is built around.
- Designed the real-time layer end to end: JWT auth with automatic token refresh, and a WebSocket event bus (FastAPI + asyncio) broadcasting live table/order state to every connected POS terminal, kitchen display, and customer display.
- Engineered a two-phase commit and startup reconciliation job to close a data-loss window on payment completion — reliability work scoped and delivered independently once the failure mode was identified in production.
- Built three companion apps from scratch, forming the control hub for the ecosystem: a customer-facing display app (live order mirroring, scan-to-pay via hosted card checkout, email receipts), a manager reporting dashboard (sales/insights analytics, staff and payment-method breakdowns), and a back-office admin hub (React/Vite/TypeScript) for menu, pricing, and margin management, including a data-migration path from a legacy POS system.
- Extended an existing kitchen display system with new real-time and recovery features: WebSocket reconnect hardening, hidden-order recall with search, and a dedicated kitchen-ticket print format.
- Contributed the POS-facing feature set to a shared FastAPI/MSSQL backend used across the ecosystem, including the combo/discount engine, seat-level billing, shift management, and the real-time event API.

Junior Developer (Part-Time) — Tecwise

August 2025 – October 2025 | Centurion

Contributed to the ongoing development of a multi-tenant Electronic Dealer Portal built on a Frappe-based ERPNext system, used by multiple automotive dealerships across South Africa. Worked closely with senior engineers to enhance both backend and frontend functionality, using Python, JavaScript, and HTML to deliver stable, user-friendly solutions aligned with business workflows.

- Implemented backend logic and frontend updates to improve usability and performance.
- Supported modular, reusable code design to ensure scalability across multiple tenants.
- Assisted in restoring and refining legacy features, ensuring compatibility with modern systems.
- Collaborated in an Agile development environment, using Git and Visual Studio Code to track progress and maintain code quality.

Data Capturer — Premium Ideas

December 2022 – January 2023

- Part of a team that migrated data from a legacy system to a new operational platform.

HIGHLIGHTED PROJECT — AI CONTEXT INFRASTRUCTURE

Designed a persistent AI context layer for a multi-repository SaaS ecosystem (8 applications, ~90 API routes), giving developers and AI agents structured, source-verifiable knowledge to reason about dependencies, business logic, and change impact.

- Layered knowledge architecture: high-level summary → service-specific reference → in-depth technical analysis, with source-based confidence ratings to cut hallucinations.
- Coordinated parallel AI research agents to cross-verify a large multi-service codebase and map cross-service dependencies.
- Result: faster feature development, more accurate AI-assisted code generation, and lower onboarding costs for new developers.

TECHNICAL SKILLS

Languages & Databases

Python, TypeScript, JavaScript, SQL, MSSQL

Frameworks & Tools

FastAPI, React Native, Vite, SQLAlchemy, AsyncStorage, Git / GitHub

Integration & Architecture

REST APIs, JWT Authentication, WebSockets, Microservices

AI & Automation

Claude AI, AI-Assisted Development, Agentic Workflows, Model Context Protocol (MCP), Prompt Engineering

EDUCATION

BSc Information Technology — North-West University

2022 – 2025

National Senior Certificate — Leeuwenhof Academy

2017 – 2021

CERTIFICATIONS

- AI Fluency: Framework & Foundations — Anthropic Academy
- AI Fluency: AI Capabilities & Limitations — Anthropic Academy
- Claude Code 101 — Anthropic Academy
- Claude Code in Action — Anthropic Academy
- Claude with the Anthropic API — Anthropic Academy
- Introduction to Model Context Protocol — Anthropic Academy
- Introduction to Agent Skills — Anthropic Academy

LANGUAGES

Afrikaans (Native) • English (Advanced)

STRENGTHS

◆ Analytical Thinking

Breaks down complex systems methodically to reach clear, actionable insights.

◆ Attention to Detail

Delivers clean, reliable work with consistent focus on accuracy.

◆ Fast Learner

Quickly picks up new tools, frameworks, and technologies.

◆ Problem-Solving

Resourceful and solution-focused when tackling technical or business challenges.

◆ Systems Thinking

Strong architectural analysis and a clear view of how components interact.

◆ Team Collaboration

Works effectively with both technical and business stakeholders.